1.WAP in C/ Java to find a root of the equation x^2-4x-10=0 using bisection method.

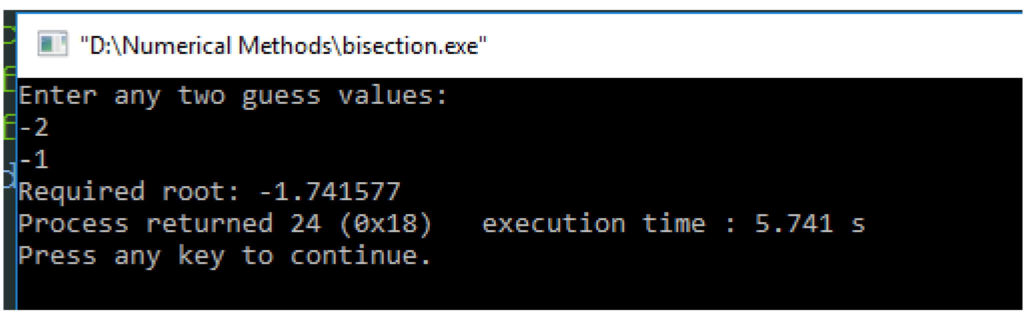
Solution:

```
#include<stdio.h>
#include<math.h>
#define f(x) (pow(x,2)-(4*x)-10)
#define E 0.0001
void main() {
    float x1,x2,x0,f0,f1,f2,root;
    printf("Enter any two guess values:\n");
    scanf("%f %f", &x1, &x2);
    f1 = f(x1);
    f2 = f(x2);
    if(f1 * f2 > 0) {
```

```
        printf("Initial guess value does not bracket the root.");
    } else {
        label:
        x0 = (x1+x2)/2;
        f0 = f(x0);
        if(f0 == 0) {
            root = x0;
        }
        if(f0 * f1 < 0) {
            x2 = x0;
            f2 = f0;
        } else {
            x1 = x0;
            f1 = f0;
        }
```

```
        if(fabs((x2-x1)/x2) < E) {
          root = x0;
        }
        else {
           goto label;
        }
        printf("Required root: %f", root);
    }
}
```

<u>Output:</u>



2.WAP in C/ Java to find a root of the equation x^2-x-2=0 using false position method in the range 1<x<3.

<u>Solution:</u>

```
#include<stdio.h>
#include<math.h>
#define f(x) (pow(x, 2)-x-2)
#define E 0.0001
void main() {
   float x1,x2,x0,f0,f1,f2,root;
   printf("Enter any two guess values:\n");
   scanf("%f %f", &x1, &x2);
   f1 = f(x1);
   f2 = f(x2);
   if(f1 * f2 > 0) {
```

```
        printf("Initial guess value does not bracket
the root.");
    } else {
        label:
        x0 = x1- f1*((x2-x1)/(f2-f1));
        f0 = f(x0);
        if(f0 == 0) {
            root = x0;
        }
        if(f0 * f1 < 0) {
            x2 = x0;
        } else {
            x1 = x0;
        }
        if(fabs((x2-x1)/x2) < E) {
           root = x0;
```

```
        }
        else {
            goto label;
        }
        printf("Required root: %f", root);
    }
}
```

Output:

```
"D:\Numerical Methods\falsePosition.exe"
Enter any two guess values:
1
3
Required root: 2.000064
Process returned 23 (0x17)   execution time : 12.244 s
Press any key to continue.
```

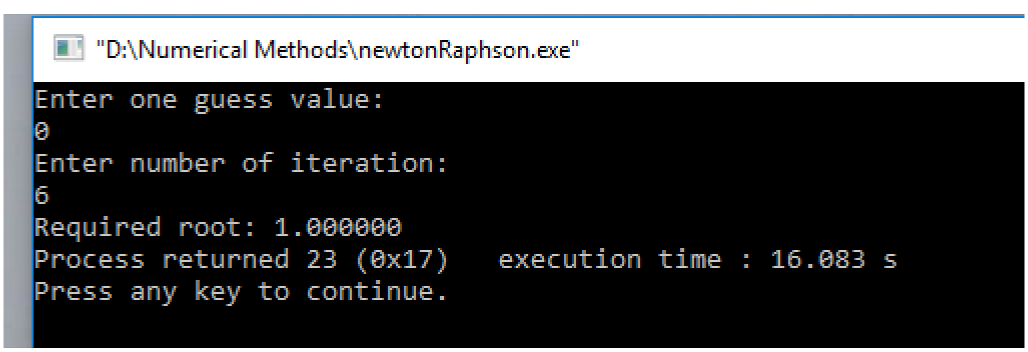
3.WAP in C/ Java to find a root of the equation x^2-3x+2=0using Newton Raphson method in the vicinity of x=0.

Solution:

```
#include<stdio.h>
#include<math.h>
#define f(x) (pow(x,2)-(3*x)+2)
#define g(x)  2*x-3
//#define E 0.0001
void main() {
  float x1,x0,f0,g0,root;
  int n, itera = 1;
  printf("Enter one guess value:\n");
  scanf("%d", &x0);
  printf("Enter number of iteration: \n");
  scanf("%d", &n);
```

```
label:
    f0 = f(x0);
    g0 = g(x0);
    x1 = x0 - (f0/g0);
    x0=x1;
    itera++;
    if(itera <= n) {
      goto label;
    } else {
      root = x1;
    }
    printf("Required root: %f",root);
}
```

<u>Output:</u>



```
"D:\Numerical Methods\newtonRaphson.exe"
Enter one guess value:
0
Enter number of iteration:
6
Required root: 1.000000
Process returned 23 (0x17)   execution time : 16.083 s
Press any key to continue.
```

4.WAP in C/ Java to find a root of the equation $x^2+x-2=0$ using the fixed point method.

Solution:

```
#include<stdio.h>
#include<math.h>
#define g(x)  2-x*x
void main() {
  float x1,x0,g0,root;
  int n, itera = 1;
  printf("Enter one guess value:\n");
  scanf("%d", &x0);
  printf("Enter number of iteration: \n");
  scanf("%d", &n);
  label:
    x1 = g(x0);
```

```
    x0 = x1;


    itera++;
    if(itera <= n) {
      goto label;
    } else {
      root = x1;
    }
    printf("Required root: %f",root);
}
```

<u>Output:</u>